Prince Choudhary

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Github | Linked-in

Skills

Language and Frameworks: Python, OpenCV, PyTorch, SQL, Scikit-learn, TensorFlow, NumPy, Pandas, Django, Flask

Algorithms: Neural Networks (ANN), CNN, YOLO, OCR, ONNX, TensorRT, LLM, Clustering, Ensemble learning

Technologies & Tools: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Data preprocessing and visualisation, AWS EC2 and S3, RAG Applications, APIs Development, Git/GitHub, Linux/Unix, Nvidia CUDA/cuDNN, Jetson Nano

Work Experience

Vinfotech, Indore Mar 2025 - Present

ML Engineer

- **Real-Time Facial Recognition & Person Tracking system:** Trained **YOLO Model** for face detection and integrated with MediaPipe to score face quality and implemented frame filtering to select the best quality faces.
- Optimized pipeline by encoding filtered faces with a **deep learning model** and sending only unknowns to AWS Rekognition, **reducing API calls by 40% as cost.**
- Quantized model to **ONNX for Jetson Orin Nano** and optimized frame processing via multithreading and larger **RTSP buffer** to prevent frame drops.

RTSP Video Streaming Pipeline: Developed a video streaming system to ingest RTSP streams over TCP from NVR cameras, **re-encoded using FFmpeg for optimized playback** and delivered via a MediaMTX server for seamless live monitoring and video analytics.

Camncloud, Indore Nov 2023 - Feb 2025

Computer Vision Engineer

- **QR/Barcode scanner on a moving conveyor belt:** **Trained YOLO model** for real-time QR/barcode detection on fast-moving objects.
- Implemented an **image processing pipeline** with techniques like Gaussian blur, adaptive thresholding, and others to **improve detection accuracy by 16%.**

GeoSyncVision: Satellite Data and Video Frame Mapping: Engineered a dual-camera setup to capture synchronized footage and construct seamless panoramic views.

- Developed a system to map geographic coordinates (latitude and longitude) from satellite data to individual video frames by timestamp alignment.
- Overlaid geographic data on each frame and converted them into a synchronized video.

BoundaryGuard: Automated Trespass Detection: Developed a trespassing detection system using YOLOv8 with line-crossing logic and polygonal ROIs via OpenCV's pointPolygonTest for accurate, real-time violation alerts.

- **Reducing system load by 20%** with an adaptive frame skipping mechanism that dynamically resumes processing.

SUREBOT, Bangalore Mar 2022 - May 2022

Web Application Developer Intern

- Developed REST APIs for HRMS application: Utilized Django REST framework and SQL for efficient data handling and backend processes.

Education

VTU Bangalore Aug 2017 - Jun 2021

B.E. in Electrical and Electronics Engineering

BSF- IT Bangalore Aug 2014 - Jun 2017

Diploma in Electrical and Electronics Engineering